

0 6 . 1

Give **two** types of cell that can stimulate an immune response.

[2 marks]

1 _____

2 _____

An autoimmune disease causes the immune system to attack healthy body tissues. Scientists investigated the immune responses of healthy mice and mice with autoimmune disease.

The chemical OXA causes an immune response in mice and can make their skin swell. Mice had olive oil applied to their left ear and OXA in olive oil applied to their right ear.

The immune response was recorded in two ways:

- the cellular response by measuring the mean increase in ear thickness 24 hours after exposure to OXA
- the humoral response by measuring the mean concentration of anti-OXA antibody in blood 14 days after exposure to OXA.

Table 2 shows the results of this investigation. The values in the brackets show ± 2 standard deviations. A value of ± 2 standard deviations from the mean includes over 95% of the data.

Table 2

Type of mice	Sex of mice	Mean increase in ear thickness / $\text{cm} \times 10^{-3}$	Mean concentration of anti-OXA antibody / arbitrary units
Healthy	Male	17.9 (± 4.1)	16 (± 3)
	Female	18.5 (± 2.9)	14 (± 4)
Autoimmune disease	Male	25.9 (± 4.5)	14 (± 2)
	Female	16.7 (± 3.0)	26 (± 7)

0 6 . 2

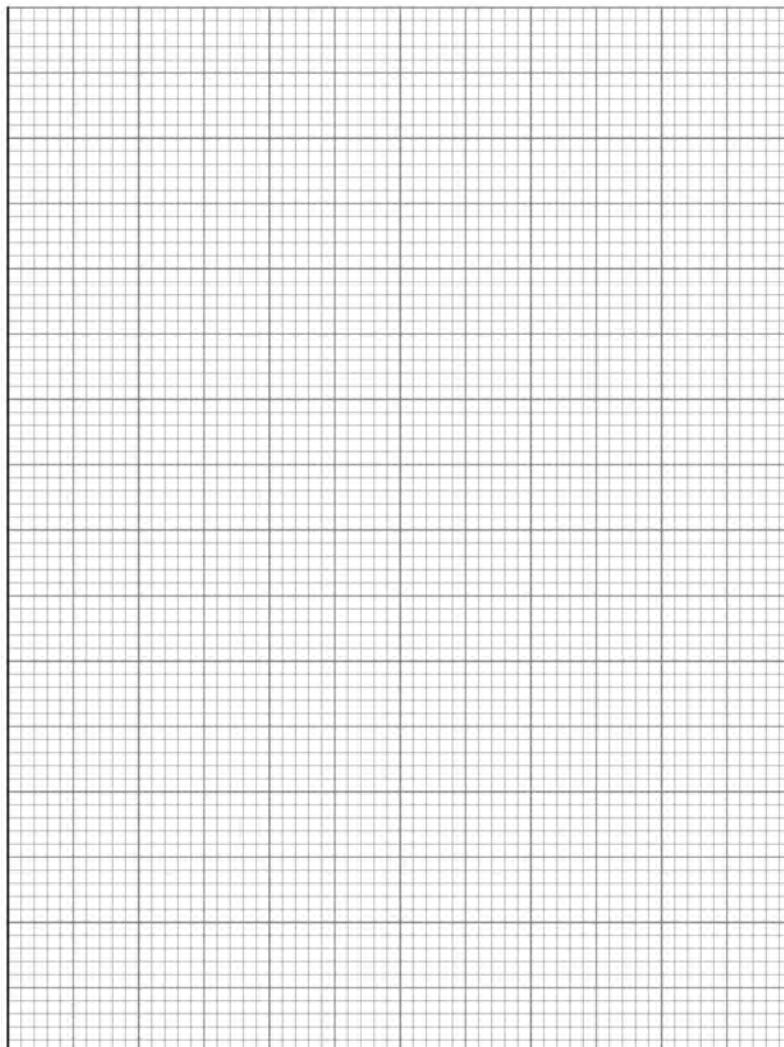
Suggest and explain **one** reason why olive oil was applied to the left ear of the mice.

[1 mark]

0 6 . 3

Plot a suitable graph for mean increase in ear thickness for each group of mice in **Table 2**.

Include the data for ± 2 standard deviations on your graph.

[3 marks]

Question 6 continues on the next page

0 6 . 4

What can you conclude about the effects of autoimmune disease on the cellular response and the humoral response in male and female mice?

Use the data to justify your conclusions.

[3 marks]

- accelerates the progression of systemic lupus erythematosus (SLE)
- prevents the progression of rheumatoid arthritis (RA).

Table 3 shows the scientists' results.

Table 3

Type of mice	Effect of oestrogen on humoral response	Effect of oestrogen on cellular response
Healthy	No effect	No effect
Autoimmune disease	Increase in response	Decrease in response

1. In humans, SLE is caused by an overproduction of antibodies.
2. In humans, RA is caused by an overproduction of cytotoxic T cells (T_C cells).

[4 marks]

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In mice, one type of autoimmune disease is inherited as a dominant allele. Would the Hardy–Weinberg principle hold true for a population of mice, some of which had this autoimmune disease?

Explain your answer.

[2 marks]

1

Section B

Answer **one** question.

You are advised to spend no more than 45 minutes on this section.